

Teacher

1.2 Classwork: Arithmetic Sequences with Variables

1. Consider the arithmetic sequence

83, 77, 71, 65, ...

- (a) Write down the value of the common difference d .
- (b) Find an expression for u_n , the n^{th} term.
- (c) Find u_{20} .
- (d) One term of the sequence is equal to -13 . Find the value of n for that term. Show your algebraic working.

(a) $d = 77 - 83 = -6$

(b) $u_n = 83 + (-6)(n-1)$

(c) $u_{20} = 83 + (-6)(20-1)$
 $= -31$

(d) $u_n = 83 - 6(n-1) = -13$
 $96 = 6(n-1)$
 $n = 17$

2. The first three terms of an arithmetic sequence are

$$u_1 = k - 4, \quad u_2 = 2k, \quad u_3 = 4k - 1.$$

- (a) Find an expression, in terms of k , for $u_2 - u_1$ and an expression for $u_3 - u_2$. Simplify your answers.
- (b) Use the fact that the sequence is arithmetic to find the value of k .
- (c) Write down the first three terms of the sequence.
- (d) Find u_{10} .

$$(a) \quad u_2 - u_1 = 2k - (k - 4) \\ = k + 4$$

$$u_3 - u_2 = (4k - 1) - 2k \\ = 2k - 1$$

$$(b) \quad k + 4 = 2k - 1 \\ k = 5$$

$$(c) \quad 1, 10, 19$$

$$(d) \quad u_{10} = 1 + \overset{9}{\cancel{8}}(10 - 1) \\ = \cancel{46} \\ = 82$$